PROJECT REPORT

CUSTOMER CHURN PREDICTION USING IBM SPSS MODELER

Submitted in partial fulfilment of the requirements for the award of the degree

BACHELOR OF BUSINESS ADMINISTRATION
INTERNATIONAL BUSINESS MANAGEMENT

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I am also thankful to United University for providing me with the opportunity to undertake this academic project. I acknowledge the learning resources and software exposure that helped me understand data preparation, classification techniques, and the interpretation of analytical outputs.

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ABSTRACT

Customer churn is an important business challenge because customers who discontinue a company's products or services can reduce revenue, increase customer acquisition costs, and affect long-term business growth. Predicting churn in advance can help organizations identify customer groups that may be at risk and plan suitable retention activities.

The main objective of this study is to demonstrate how predictive analytics can be applied to customer churn prediction using IBM SPSS Modeler. For academic practice, a synthetic dataset containing 100 customer records and 18 attributes has been prepared. The attributes include customer tenure, contract type, internet service, payment method, monthly charges, support tickets, and other customer characteristics. The target variable, Churn, has two categories: Yes and No.

The project explains the stages of data import, data inspection, variable-role assignment, data exploration, classification model development, and model evaluation. A decision-tree method such as CHAID or C5.0 may be selected depending on the algorithms available in the installed IBM SPSS Modeler version. The final algorithm and its outputs must be confirmed after execution in the software.

In the prepared sample, 38 records are labelled as churn and 62 as non-churn. This represents a sample churn proportion of 38.0%. These figures describe only the synthetic teaching dataset and should not be interpreted as an industry churn rate. Model accuracy and other evaluation measures are intentionally not stated in advance; they must be recorded from the actual IBM SPSS Modeler output.

Keywords: Customer Churn, Customer Retention, Predictive Analytics, IBM SPSS Modeler, Decision Tree, Marketing Analytics.

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Note: Update page numbers after final formatting and after inserting the actual IBM SPSS Modeler screenshots and outputs.

1. INTRODUCTION

1.1.1 Meaning of Churn Prediction

Churn prediction is the process of using customer information and analytical techniques to estimate whether a customer is likely to stop using a company's product or service. The purpose is not simply to identify customers who have already left. It is to recognize patterns that may indicate a risk of leaving so that the business can investigate and take suitable action before the customer discontinues the service.

1.1.2 Why Churn Prediction Is Important

Acquiring a new customer often requires spending on advertising, sales activities, introductory offers, and onboarding. When an existing customer leaves, the organization may lose future purchases or recurring subscription revenue and may need to spend again to attract a replacement. Churn prediction can help a business focus attention on customer experience, service quality, pricing, and relationship

management. It can also help teams plan retention activities more systematically.

1.1.3 How Customer Churn Can Occur

Churn may be voluntary or involuntary. Voluntary churn occurs when a customer chooses to leave, for example because of price, dissatisfaction, a competitor's offer, or a change in personal needs. Involuntary churn may occur because of payment failure, account closure, or another administrative reason. The present teaching dataset uses a single Churn field with Yes and No categories; it does not separately identify voluntary and involuntary churn.

1.1.4 Common Factors That May Be Related to Churn

Customer churn can be associated with several types of information. Tenure describes how long a customer has remained with a provider. Contract type describes the length or flexibility of the agreement. Monthly charges and payment method describe aspects of billing. Service fields describe the products or support options used by a customer. Support-ticket counts may indicate that a customer has contacted the provider for help. These fields can be examined for patterns, but a relationship in the data does not automatically prove that a field causes churn.

1.1.5 How a Prediction Model Works

A classification model learns patterns from records for which the outcome is already known. In this project, the known outcome is the Churn field. The model examines input attributes and creates rules or splits that separate records into groups with different observed churn outcomes. After the model is developed, it can assign a predicted class or estimated probability to records that are being assessed. The quality of those predictions must be checked using data that was not used to train the model.

1.1.6 Example of Business Use

For example, a telecom provider may notice that some customers have short tenure, month-to-month contracts, higher monthly charges, or repeated support contacts. A churn model may identify combinations of these characteristics that are associated with a higher churn rate in the training data. The retention team can then review the customer's circumstances, check whether there is a service issue, and decide whether a suitable support response or plan option is appropriate. The model should guide investigation rather than make the decision automatically.

1.1.7 Meaning of the Churn Target in This Project

The target field in this project is named Churn. A value of Yes represents a customer record labelled as churned in the synthetic dataset, while No represents a record labelled as not churned. All other relevant fields are potential input variables. CustomerID is only a record identifier and should not be treated as a meaningful predictor.

1.1 Background of the Study

Customer churn refers to a situation in which a customer stops purchasing from a business, cancels a subscription, or discontinues a service. It is a major concern for subscription-based businesses such as telecommunications, internet service providers, banking, insurance, and digital platforms. When customers leave, organizations may lose recurring revenue and may need to spend additional money on attracting replacement customers.

In competitive markets, customers can compare prices, service quality, product features, and offers from different providers. Poor customer experience, high charges, unsuitable plans, billing issues, weak customer support, and attractive competitor offers may influence a customer's decision to leave.

1.2 Predictive Analytics in Customer Retention

Predictive analytics uses existing data, statistical methods, and machine-learning techniques to identify patterns and estimate possible future outcomes. In customer relationship management, predictive analytics can help businesses identify customer groups that show characteristics associated with churn. These insights may support retention campaigns, service improvements, personalized offers, and better customer communication.

1.3 Role of IBM SPSS Modeler

IBM SPSS Modeler is a data-mining and predictive-analytics application. It provides a visual workflow for importing data, preparing fields, exploring patterns, building models, and examining model outputs. In this project, the software is intended to demonstrate a classification process in which customer attributes are used as input fields and Churn is treated as the target field.

1.4 Overview of the Present Project

The present project uses a synthetic sample of 100 customer records and 18 attributes. The dataset is designed for learning and demonstration only. It does not represent customers of an actual organization. The project focuses on understanding the data, assigning the correct field roles, selecting a suitable classification technique, and interpreting the output carefully.

1.5 Importance of the Study

It introduces the concept of customer churn and its effect on business performance.

It demonstrates how customer information can be prepared for predictive analysis.

It connects marketing analytics with customer relationship management.

It explains how prediction can support customer-retention planning.

It provides practical exposure to IBM SPSS Modeler and classification concepts.

2. PROBLEM STATEMENT

Businesses operating in competitive markets face the risk of losing customers to alternative providers. Customer churn can affect recurring revenue, customer lifetime value, market share, and the cost of acquiring new customers. Organizations therefore need to understand the factors that may be associated with customers discontinuing a product or service.

Traditional reporting may show how many customers have already left, but it may not always help identify patterns among customers who are more likely to leave in the future. Predictive analytics can be used to examine customer characteristics and classify records into churn and non-churn categories.

The main problem addressed in this project is to demonstrate how customer-related attributes—such as tenure, contract type, monthly charges, payment method, service subscriptions, and support-ticket history—can be analysed to understand their relationship with customer churn.

The project uses IBM SPSS Modeler to prepare a synthetic customer dataset and develop a classification model. The target variable is Churn, with two categories: Yes and No. The model's performance must be assessed using actual software outputs and appropriate validation measures before any claim about predictive reliability is made.

2.1 Research Questions

What is customer churn and why is it important for businesses?

Which customer attributes are available for studying churn in the sample dataset?

How can IBM SPSS Modeler be used to prepare customer data and build a classification model?

How can model output support customer-retention planning?

What limitations should be considered when interpreting results from a small synthetic dataset?

3. DATA DISCUSSION

3.1 Overview of the Dataset

The prepared dataset contains 100 customer records and 18 attributes. Each row represents one synthetic customer record. The dataset includes demographic indicators, service information, contract details, billing information, customer tenure, and the churn outcome.

3.2 Source of Data

The data is synthetic and has been generated for academic practice. It does not represent real customers, an actual company, or verified market behaviour. The values are intended to provide a structured dataset for learning the steps involved in data preparation and classification.

3.3 Description of Important Variables

Variable	Description
CustomerID	Unique customer code; use as an identifier, not a predictor.
Gender	Customer gender category in the synthetic sample.
SeniorCitizen	Indicates whether the customer belongs to the senior-citizen group.
Partner	Indicates whether the customer has a partner.
Dependents	Indicates whether the customer has dependents.
TenureMonths	Number of months the customer has remained with the service.
PhoneService	Indicates whether phone service is subscribed.
InternetService	Internet service category: DSL, Fiber optic, or No.
Contract	Contract duration: Month-to-month, One year, or Two year.
PaperlessBilling	Indicates whether paperless billing is enabled.
PaymentMethod	Customer's payment method.

Variable	Description
MonthlyCharges	Illustrative monthly service charges.
TotalCharges	Illustrative accumulated charges.
TechSupport	Technical-support subscription status.
OnlineSecurity	Online-security subscription status.
StreamingTV	Indicates whether streaming TV is subscribed.
SupportTickets	Number of support tickets in the sample period.
Churn	Target variable: Yes or No.

3.4 Target Variable

The target variable is Churn. It contains two categories: Yes and No. A value of Yes indicates that the customer is labelled as having discontinued the service in the synthetic sample. A value of No indicates that the customer is labelled as continuing the service. The remaining relevant fields are potential input variables.

3.5 Dataset Summary

Churn Category	Number of Records	Percentage
No	62	62.0%
Yes	38	38.0%
Total	100	100.0%

3.6 Preliminary Descriptive Observations

The average tenure in the churn-labelled group is approximately 22.2 months, compared with 47.2 months in the non-churn group. The average monthly charge is approximately 64.08 in the churn-labelled group and 57.70 in the non-churn group. These are descriptive summaries of this generated sample only; they do not establish causation or prove that any variable independently causes churn.

3.7 Churn by Contract Type

Contract Type	No Churn	Churn	Churn within Contract Group
Month-to-month	19	31	62.0%
One year	22	7	24.1%
Two year	21	0	0.0%

The contract comparison is included to demonstrate how categorical customer attributes can be explored. Any pattern in this small synthetic dataset should be treated as an educational observation and not as a general conclusion about real customers.

4. METHODS, TOOLS AND TECHNOLOGIES USED

4.1 Predictive Analytics

Predictive analytics is a data-analysis approach that uses historical or existing data to estimate possible outcomes. In this project, the method is applied to customer information to classify the churn outcome as Yes or No.

4.2 IBM SPSS Modeler

IBM SPSS Modeler is used for data import, field inspection, data preparation, model development, and output interpretation. Its visual stream-based interface allows users to connect nodes and create a sequence of analytical steps.

4.3 Dataset and Variables

The project uses the synthetic dataset supplied with this report. It contains 100 records and 18 attributes. CustomerID should be treated as an identifier and should not normally be used as a predictor because it is a record code rather than a meaningful customer behaviour measure.

4.4 Data Preparation

Import the Excel or CSV file into IBM SPSS Modeler.

Inspect field names, data types, measurement levels, and missing values.

Set CustomerID as an identifier or exclude it from predictor fields.

Set Churn as the target field with values Yes and No.

Set the remaining relevant attributes as input fields.

Check that numeric variables such as TenureMonths, MonthlyCharges, TotalCharges, and SupportTickets have suitable measurement levels.

Confirm that categorical fields are recognized as nominal or ordinal as appropriate.

4.5 Classification Algorithms

A decision-tree algorithm can be used for this binary classification task. CHAID (Chi-squared Automatic Interaction Detection) creates splits based on relationships between input fields and the target. C5.0 is another classification-tree method available in some IBM SPSS Modeler installations. The final algorithm should be selected according to the installed software version and the project supervisor's requirements. The report must name the algorithm that was actually run.

In a decision tree, the data is divided into smaller groups according to the input field that best separates the target categories under the selected algorithm's criteria. The final tree can be read as a set of decision rules. For example, a branch may divide customers by contract type and then examine another field within one of those groups. The exact splits must be taken from the model actually produced by the software; they should not be written in advance.

4.6 Model Evaluation Techniques

Measure	Meaning
Accuracy	Proportion of all records classified correctly.
Precision	Among records predicted as churn, the proportion that are actually churn.
Recall	Among actual churn records, the proportion correctly identified.
Confusion Matrix	Table comparing actual and predicted classes.
Validation / Test Sample	Records kept separate from model training to examine performance on unseen cases.

No accuracy, precision, recall, or confusion-matrix values are assumed in this report. These values must be inserted only after the model has been executed and evaluated in IBM SPSS Modeler.

5. WORKFLOW

The workflow consists of a sequence of steps from importing the customer dataset to evaluating the classification model. The steps below provide a practical guide for implementing the project in IBM SPSS Modeler.

Step	Stage	Activity
1	Dataset Import	Load the prepared Excel or CSV customer dataset.
2	Data Inspection	Check field names, types, measurement levels, and missing values.
3	Type / Field Roles	Set Churn as Target, CustomerID as Record ID, and suitable fields as Inputs.
4	Data Exploration	Review churn counts and compare churn across customer attributes.
5	Partition (if available)	Separate training and testing data for evaluation.
6	Model Building	Run a suitable classification tree such as CHAID or C5.0.
7	Prediction	Generate predicted churn classes and, where available, probabilities.
8	Evaluation	Review confusion matrix and performance measures on appropriate test data.
9	Interpretation	Explain important splits and possible retention implications cautiously.

5.1 Workflow Diagram

Customer Dataset → Import → Inspect & Prepare → Assign Field Roles
↓
Explore Data → Partition Data → Build Classification Model
↓
Generate Predictions → Evaluate Model → Findings & Recommendations

5.2 Practical Note

The exact nodes and node names may differ according to the installed version of IBM SPSS Modeler. Screenshots should be captured from the student's own software session after completing each stage.

6. OBJECTIVES OF THE STUDY

To understand the concept of customer churn and its importance in business.

To examine customer-related attributes that may be associated with churn.

To prepare and inspect a synthetic customer dataset using IBM SPSS Modeler.

To apply a suitable classification technique to the Churn target variable.

To understand how model outputs can support customer-retention planning.

To identify the limitations of using a small synthetic dataset for predictive analysis.

7. SCOPE AND SIGNIFICANCE OF THE STUDY

7.1 Scope of the Study

The scope of this study is limited to demonstrating the application of predictive analytics to customer churn using IBM SPSS Modeler. It focuses on a synthetic dataset of 100 customer records and 18 attributes. The project covers data import, data inspection, variable-role assignment, exploratory summaries, classification model development, and interpretation of available model output.

Understanding customer churn and customer retention.

Exploring customer tenure, contract, service, billing, and support attributes.

Preparing customer data for classification in IBM SPSS Modeler.

Applying a decision-tree classification method, subject to software availability.

Understanding model evaluation and the responsible use of predictions.

7.2 Significance of the Study

Marketing decision support: churn analysis can help marketing teams think about customer-retention activities.

Customer relationship management: analysis can help identify customer groups for further review.

Service improvement: patterns in support tickets and service subscriptions can guide questions for service teams.

Practical learning: the project provides experience with data preparation and classification.

Analytical awareness: it demonstrates why model validation and data quality matter.

The study is educational. It does not provide a validated churn scoring system for a real business. Actual use would require suitable company data, privacy safeguards, model validation, and review by business specialists.

8. MODEL OUTPUT AND RESULTS

8.1 Introduction

This chapter is designed to present the output obtained after running the classification model in IBM SPSS Modeler. The descriptive dataset figures below are calculated from the supplied synthetic file. The model-specific outputs must be completed from the actual software run.

8.2 Dataset Summary

Category	Records	Share
Churn = No	62	62.0%
Churn = Yes	38	38.0%
Total	100	100.0%

8.3 Preliminary Descriptive Findings

The dataset contains 100 customer records and 18 attributes.

38 records are labelled Churn = Yes and 62 are labelled Churn = No.

The sample churn proportion is 38.0%; this is a property of the synthetic sample, not an industry benchmark.

Average tenure is 22.2 months for churn-labelled records and 47.2 months for non-churn records.

Contract type, service type, billing, payment method, tenure, and support history can be explored as potential predictors.

8.4 IBM SPSS Model Output – To Be Inserted After Execution

Insert the actual model nugget / tree output screenshot here. Record the selected algorithm, number of records used, tree depth, number of nodes, important predictor fields, and the model's target field exactly as shown by IBM SPSS Modeler.

Model Output Item	Actual Value from IBM SPSS Modeler
Algorithm selected	[Enter actual algorithm]
Target field	Churn
Training records	[Enter software output]
Testing / validation records	[Enter software output]
Tree depth / number of nodes	[Enter if applicable]
Important predictors / splits	[Enter from actual output]
Accuracy	[Enter only if generated]
Precision / Recall	[Enter only if generated]

8.5 Confusion Matrix – To Be Inserted

Insert the confusion matrix produced from the test or evaluation sample. Explain the number of correctly and incorrectly classified records in each class. Do not use training performance as a substitute for independent test performance.

Actual / Predicted	Predicted No	Predicted Yes
Actual No	[TN]	[FP]
Actual Yes	[FN]	[TP]

8.6 Interpretation of Results

Interpretation should be based on the actual tree splits and evaluation results. If a variable appears as an important split, describe the groups formed by the model and the observed churn distribution in those groups. Avoid claiming that a variable causes churn. A predictive association in a small synthetic dataset is not proof of a causal relationship.

8.7 Conclusion of Model Output

The final conclusion for this chapter should be written after the IBM SPSS Modeler run. It should summarize the model structure, the evaluation results, and the main limitations. Until then, the current report provides the dataset summary and a structured space for recording the actual software output.

9. RECOMMENDATIONS AND SUGGESTIONS

Use churn analysis as a decision-support tool: businesses can examine customer patterns and use the findings to plan further customer research.

Improve customer experience: monitor service quality, complaint handling, billing clarity, and response time.

Review contract and plan suitability: offer plans that match customer needs and usage patterns rather than relying only on discounts.

Strengthen customer support: review repeated support issues and resolve recurring service problems.

Develop responsible retention campaigns: tailor communication to customer needs and avoid excessive or irrelevant promotional messages.

Maintain data quality: check missing values, inconsistent categories, duplicates, and incorrect entries before model building.

Validate model performance: use a separate test sample and review accuracy, precision, recall, and the confusion matrix where available.

Protect customer privacy: restrict access to personal information and use data only for appropriate, disclosed purposes.

Avoid automated adverse decisions: model predictions should not be the sole basis for denying service or making other significant customer decisions.

Improve future research: use larger, authentic, representative, and appropriately consented datasets for real-world study.

Overall Recommendation

Predictive analytics may help organizations understand customer churn patterns and consider suitable retention strategies. However, model predictions should be combined with customer feedback, service knowledge, and professional judgment. A model trained on synthetic data should not be used to make decisions about real customers.

10. LIMITATIONS OF THE STUDY

Small dataset: the project uses only 100 customer records, which is insufficient to represent the diversity of customers in a large business.

Synthetic data: the records are generated for academic practice and do not reflect actual customer behaviour.

Limited variables: important factors such as customer satisfaction surveys, competitor activity, customer lifetime value, and detailed complaint history are not included.

Model validation: predictive performance cannot be established until the model is run and evaluated on appropriate test data.

Potential sample bias: the sample distribution and relationships are generated for demonstration and may not reflect real-world churn patterns.

Limited generalizability: findings from this sample cannot be generalized to all industries or organizations.

Software-version differences: available algorithms, nodes, and output formats may vary across IBM SPSS Modeler versions.

Privacy and ethics: a real implementation would require lawful data collection, appropriate access controls, and responsible use of customer information.

Overall Limitation

Despite these limitations, the project provides a basic understanding of customer churn, data preparation, and predictive modelling. Further research using larger and authentic datasets, relevant variables, and proper validation could provide more reliable insights.

11. CONCLUSION

The project titled “Customer Churn Prediction Using IBM SPSS Modeler” focuses on the application of predictive analytics to a common business challenge: customers discontinuing products or services. Customer churn can affect recurring revenue, customer relationships, and the cost of acquiring new customers. Understanding customer patterns can therefore support better customer-retention planning.

A synthetic dataset containing 100 customer records and 18 attributes was prepared for this study. The target variable, Churn, contains 38 Yes records and 62 No records. The dataset includes customer tenure, contract type, service information, billing details, payment method, and support-ticket information. These attributes provide a basis for demonstrating data exploration and classification.

The project describes how IBM SPSS Modeler can be used to import data, inspect fields, assign variable roles, explore customer information, build a classification model, and evaluate its output. A decision-tree algorithm such as CHAID or C5.0 may be used depending on the installed version and the actual project execution.

The descriptive figures in this report are calculated from the synthetic dataset. Model accuracy and other predictive performance results have not been assumed. They must be added after the model is executed and evaluated in IBM SPSS Modeler. This distinction is important because a model's structure alone does not prove that it predicts churn accurately.

In conclusion, predictive analytics can serve as a decision-support approach for understanding customer churn and considering retention strategies. Future studies using larger, authentic, representative datasets and validated models could provide more dependable insights for business decision-making.

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Note: Verify the exact edition and publication details of books and software documentation consulted before final submission.

APPENDIX A: DATASET FIELD DICTIONARY

Field	Description	Role	Measurement
CustomerID	Unique customer reference	Identifier	Nominal
Gender	Customer gender category in this synthetic sample	Input	Nominal

Field	Description	Role	Measurement
SeniorCitizen	Whether customer is in the senior-citizen group	Input	Nominal
Partner	Whether customer has a partner	Input	Nominal
Dependents	Whether customer has dependents	Input	Nominal
TenureMonths	Number of months the customer has stayed	Input	Continuous
PhoneService	Whether phone service is subscribed	Input	Nominal
InternetService	Type of internet service	Input	Nominal
Contract	Customer contract duration	Input	Nominal
PaperlessBilling	Whether paperless billing is enabled	Input	Nominal
PaymentMethod	Method used to pay bills	Input	Nominal
MonthlyCharges	Current monthly service charges	Input	Continuous
TotalCharges	Approximate accumulated charges	Input	Continuous
TechSupport	Whether technical support is subscribed	Input	Nominal
OnlineSecurity	Whether online security is subscribed	Input	Nominal
StreamingTV	Whether streaming TV is subscribed	Input	Nominal
SupportTickets	Number of support tickets in the sample period	Input	Ordinal/Count
Churn	Whether the customer discontinued service	Target	Nominal

APPENDIX B: DATASET UPLOAD AND IMPORT GUIDE

1. Download customer_churn_synthetic_dataset.xlsx to your computer.

2. Open IBM SPSS Modeler and create a new stream.

3. From the Sources palette, add an Excel node (the exact node name may vary by version).

4. Open the node settings and browse to customer_churn_synthetic_dataset.xlsx.

5. Select the Customer_Data worksheet, not the Data_Dictionary or Project_Notes worksheet.

6. Confirm that the first row contains field names and preview the records.

7. Execute or preview the source node to check that the data loads correctly.

8. Add a Type node. Set Churn to Target, CustomerID to Record ID or None, and suitable explanatory fields to Input.

9. Check that numeric fields are recognized as numeric and categorical fields as nominal.

10. Save the stream so that the import settings and workflow can be reopened.

APPENDIX B: IBM SPSS MODELER EXECUTION CHECKLIST

Download and extract the supplied Excel dataset.

Open IBM SPSS Modeler and create a new stream.

Add an Excel or Var. File node and browse to the dataset.

Preview the data and confirm that all 100 records and 18 fields are visible.

Add a Type node and set Churn as Target.

Set CustomerID as Record ID or exclude it from predictors.

Set suitable customer attributes as Input and check measurement levels.

Explore the target distribution and relevant input fields.

If required, create a training/testing partition using a reproducible method.

Add and configure the selected classification model (for example CHAID or C5.0 if available).

Run the stream and inspect the model nugget and generated output.

Evaluate on test data and capture the confusion matrix and available performance measures.

Save screenshots of the stream, Type node settings, model settings, model output, and evaluation output.

Replace all bracketed placeholders in Chapter 8 with the actual software results.

APPENDIX C: PROJECT FILES

customer_churn_synthetic_dataset.xlsx – includes Customer_Data, Data_Dictionary, and Project_Notes sheets.

customer_churn_synthetic_dataset.csv – flat data file for importing into IBM SPSS Modeler.

Customer_Churn_Project_Report.docx – editable report.

Customer_Churn_Project_Report.pdf – printable report.